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Editor-in-Chief
IEEE Signal Processing Letters

Dear Editor,

I am resubmitting the manuscript entitled

"When Does Temporal Memory Help? Selective SSM vs. MLP in RL-Based Multi-Target DOA Track Management"

for consideration for publication in *IEEE Signal Processing Letters* as an original Letter.

Resubmission disclosure. This manuscript was previously assigned ID **SPL-46772-2026** and was *withdrawn by the author on 24 April 2026, prior to assignment of any reviewers and before any review reports were generated.* No reviewer feedback was received. The withdrawal was made solely to correct three clerical errors in the reference list that I identified after submission. A separate supporting document (`manuscript_history.pdf`) accompanying this resubmission provides a full disclosure and confirms compliance with IEEE Signal Processing Society Policies and Procedures Manual Section 6.16, whose "one resubmission" limit applies to manuscripts *rejected after review* and is therefore not invoked by this pre-review withdrawal.

Submission history. This is the second resubmission after two prior submissions (SPL-46772-2026 and SPL-47058-2026) were rescinded by the editorial system due to GROBID-based reference verification: the extractor flagged 61% (first submission) and 50% (second submission) of references as DOI-unverifiable. The present resubmission addresses the root cause directly: every reference now carries a verbatim \url{https://doi.org/...} string in the printed bibliography that GROBID can extract as plain text.

What changed since SPL-46772-2026 / SPL-47058-2026. The previous submissions triggered a *Research Integrity Alert* from the IEEE editorial system, with the following message returned by the GROBID reference extractor:

> *"Your manuscript has triggered a 'Research Integrity Alert'. The following reference titles cannot be verified by DOI, representing **61%** of the submission's references"*

The alert listed eleven references whose printed entries did not carry a verifiable DOI:

1. J. H. Choi. "Underdetermined direction of arrival estimation using higher-order cumulants"
2. J. H. Choi. "COP-RFS: Closed-loop underdetermined DOA estimation ...," IEEE Trans. Signal Processing
3. M. Hausknecht, P. Stone. "Deep recurrent Q-learning for partially observable MDPs"
4. S. Kapturowski et al. "Recurrent experience replay in distributed reinforcement learning"
5. N. Dorka. "Generalization, mayhems and limits in recurrent proximal policy optimization"
6. A. Gu, K. Goel, C. Ré. "Efficiently modeling long sequences with structured state spaces"
7. J. T. H. Smith et al. "Simplified state space layers for sequence modeling"

8. A. Gu, T. Dao. "Mamba: Linear-time sequence modeling with selective state spaces"
9. T. Ota. "Decision Mamba: Reinforcement learning via hybrid selective sequence modeling"
10. J. Han et al. "Audio Mamba: Bidirectional state space model for audio representation learning," IEEE SPL
11. J. Schulman et al. "Proximal policy optimization algorithms"

This was caused by IEEEtran.bst suppressing the `doi` field by default in the printed bibliography. The withdrawal allowed me to make two corrections that together address every entry on the GROBID list:

(A) Three clerical bibliographic errors:

- **Ref [8]** (arXiv:2205.11104) – author list corrected from "N. Dorka" to the actual authors "M. Pleines, M. Pallasch, F. Zimmer, and M. Preuss."
- **Ref [11]** (Ota, Decision Mamba) – title and arXiv ID corrected from "…Hybrid Selective Sequence Modeling, arXiv:2406.00079" to the actual Ota paper "Decision Mamba: Reinforcement Learning via Sequence Modeling with Selective State Spaces, arXiv:2403.19925" (arXiv:2406.00079 is a different paper by Huang et al. previously conflated with Ota's work).
- **Ref [12]** (Erol et al., Audio Mamba) – title corrected to the published IEEE SPL title "*Bidirectional* State Space Model for Audio Representation Learning" and DOI updated to 10.1109/LSP.2024.3483009 (IEEE Xplore record 10720871).

(B) DOI / verifiable identifier added to every reference and ***rendered as plain-text `https://doi.org/...` URL in the printed bibliography***. A `doi` field was added to all sixteen entries in the `.bib` file. More importantly, after SPL-47058-2026 was rescinded with 50% references still flagged, the rendering format was changed from a hyperlinked DOI string to a verbatim `\url{https://doi.org/...}` that displays the full URL as plain text in the printed bibliography. This matches the format requested in the editorial system's example ("`\url{https://doi.org/10.1177/...}`") and is directly extractable by GROBID. All eleven references named in the GROBID alert above now carry a printed DOI:

- Choi 2015 (Ref [1]): `10.1109/TSP.2014.2385036`
- Choi 2026 (Ref [3], companion under review at IEEE TSP): preprint URL provided in lieu of DOI
- Hausknecht 2015 (Ref [7]): `10.48550/arXiv.1507.06527`
- Kapturowski 2019 (no longer cited; removed from final list)
- Pleines 2022 (Ref [8], replacing the previously mis-attributed "N. Dorka" entry): `10.48550/arXiv.2205.11104`
- Gu & Dao Mamba (Ref [9]): `10.48550/arXiv.2312.00752`
- Gu, Goel, Ré S4 (Ref [10]): `10.48550/arXiv.2111.00396`
- Smith et al. S5 (no longer cited; removed)
- Ota Decision Mamba (Ref [11], with corrected arXiv ID 2403.19925): `10.48550/arXiv.2403.19925`
- Erol et al. Audio Mamba (Ref [12], IEEE SPL): `10.1109/LSP.2024.3483009`
- Schulman PPO (Ref [16]): `10.48550/arXiv.1707.06347`

The five additional references not on the GROBID list were also verified end-to-end (Vo & Ma 2006, Rosello & Kochenderfer 2018, Stephan et al. 2022, Ewers et al. 2025, Cover & Thomas 2006, Schäfer & Zimmermann 2007 with year/DOI corrected, Kaelbling et al. 1998). After this change, GROBID-style automated tools can verify all references end-to-end.

The technical content – Proposition 1, Algorithm 1, all theoretical derivations, experimental results, figures, and tables – is **unchanged** from the withdrawn version. The corrections are purely bibliographic / metadata.

Summary of contribution. This Letter addresses a fundamental question in reinforcement-learning (RL)-based adaptive multi-target DOA tracking: when does adding temporal memory to an RL policy improve performance over a memoryless multilayer perceptron? Using the data-processing inequality and the notion of a sufficient statistic, we prove that a selective state-space model (SSM) encoder outperforms an MLP if and only if the instantaneous observation is not a sufficient statistic for the policy-relevant history (Proposition 1). We instantiate this result as **Mamba-COP-RL**, coupling a Mamba SSM encoder with a PPO actor–critic operating on raw 181-dimensional COP spatial spectra. On **CombatTrackEnv**, an open-source benchmark with four combat-realistic multi-target scenarios (Jamming, Stealth, Saturation, Formation), Mamba-COP-RL improves GOSPA by up to 25.2% over fixed GM-PHD thresholds and 6.7% over MLP-PPO, with gain magnitudes ordered by temporal-dependency horizon exactly as predicted by the theory. Ablations against GRU-PPO and LSTM-PPO confirm the role of Mamba's input-dependent selectivity. The 41.4 KB INT8 model runs at 3.4 ms/scan on an STM32H7 Cortex-M7, providing a concrete edge-deployable artifact.

Scope. The submission is squarely within the scope of *IEEE Signal Processing Letters*: (i) sensor-array signal processing (higher-order-cumulant DOA estimation), (ii) statistical signal processing for multi-target tracking (random-finite-set / GM-PHD filtering), and (iii) machine learning for signal processing (selective SSMs, RL on signal-domain observations). The principal contribution is a single, sharply stated theoretical condition with a tightly scoped empirical validation – a natural fit for the Letter format.

Related submission cited as Ref [3]. A companion manuscript "*Underdetermined High-Resolution DOA Estimation and Multi-Target Tracking via COP-RFS*" (J. H. Choi, 2026) is currently under review at *IEEE Transactions on Signal Processing* and is cited as Ref [3] in this Letter. That work introduces the COP-RFS signal-processing pipeline; the present Letter adds an RL-based GM-PHD parameter scheduler on top of that pipeline and formally characterizes when temporal memory is beneficial. The two contributions are complementary and non-overlapping in their results, figures, and tables.

Code availability. A reproducible release accompanies this submission. Reviewers can reproduce the headline Table II numbers in approximately 30 seconds on a CPU laptop via a single command, `python reproduce_spl.py`. The repository (including the manuscript PDF placed alongside the reproduction script) is publicly available at `https://github.com/DrJinHoChoi/IronDome-DOA-Tracking` under a dual academic / commercial license. The exact submission commit is git-tagged `spl2026-v1`.

Author contributions and conflicts of interest. This is a sole-authored manuscript; the author conceived the proposition, designed the architecture, implemented the CombatTrackEnv benchmark and Mamba-COP-RL, ran all experiments, and wrote the manuscript. There are no conflicts of interest. No funding was received that requires disclosure.

Declarations. Apart from the previously withdrawn SPL-46772-2026 documented above, the manuscript has not been published previously, is not under review at any other journal or conference, and is not derived from any prior conference publication. No formal preprint has been posted to arXiv, TechRxiv, or any dedicated preprint server; a copy of the PDF is included in the public code repository above for reproducibility (IEEE-permissible self-hosting alongside the reproduction code).

Thank you for considering this resubmission. I am happy to provide any additional clarification the editorial board may require.

Sincerely,

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